

likelihood ratio test for circle of N cooperators/defectors

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clear; clc;

% vector of 0's and 1's
% 1 = cooperate
% 0 = defect
x = [ 1 1 0 1 1 1 0 1 0 1 0 0 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 1 1 1 1 1 1 1];

k=6; % number of neighbors (can change)
N=length(x); % total number of players

if rem(k,2)~=0 || k<2 || k>(N-1)/2
    error('Value of k is odd or out of range.')
end

%x=x(randperm(N)); % scramble players (for testing purposes)

c = sum(x); % number of cooperators
s=0;
for i=1:round(k/2);
    left = [i+1:N 1:i];
    right = [N-i+1:N 1:N-i];
    s=s+length(find(x == x(left))) + length(find(x == x(right)));
end
s=s/2;

disp([ num2str(N) ' players with states: ' num2str(x)])
disp(['cooperators: ' num2str(c) ' out of ' num2str(N)])
disp(['neighbors who agree: ' num2str(s) ' out of ' num2str(N*k/2) ])

logL1 = log(binopdf(s,N*k/2,s/(N*k/2)));

q=(c/N)^2+(1-c/N)^2;
logL0 = log(binopdf(s,N*k/2,q));
W=-2*(logL0-logL1);

alpha = 0.05; % significance level
chi2 = chi2inv(1-alpha,1);

disp(['Probability that random pair would agree'])
disp(['    given observed number of cooperators is ' num2str(q) ])
disp(['Expected number of agreeing neighbors is ' num2str(N*k*q/2) ' out of ' num2str(N*k/2)])
disp(['log(L0/L1) = ' num2str(logL0-logL1) ])
disp(['test statistic W = ' num2str(W) ])
disp(['using alpha = ' num2str(alpha) ' threshold is chi2 = ' num2str(chi2) ])

if W>chi2
    disp(['reject H0 in favor of H1'])
    % note p-value is probability of observing clustering
    % that is equal to or more extreme than current configuration
    P = 1-idchi2(W,1);
    disp(['p-value (indirect): ' num2str(P) ])
    if s>N*k*q/2
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        P2 = sum(binopdf(s:N*k/2,N*k/2,q));
    else
        P2 = sum(binopdf(0:s,N*k/2,q));
    end
    disp(['p-value    (direct): ' num2str(P2) ])

else
    disp(['cannot reject H0'])
end

return

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37 players with states: 1  1  0  1  1  1  0  1  0  1  0  0  1  1  1  1  1  1  1  1  1  1  0
    0  0  0  0  0  0  1  1  1  1  1  1  1
cooperators: 25 out of 37
neighbors who agree: 77 out of 111
Probability that random pair would agree
    given observed number of cooperators is 0.56172
Expected number of agreeing neighbors is 62.3514 out of 111
log(L0/L1) = -4.0676
test statistic W = 8.1352
using alpha = 0.05 threshold is chi2 = 3.8415
reject H0 in favor of H1
p-value (indirect): 0.0043415
p-value    (direct): 0.003013

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